



PIZZO

A Gamified Chinese Postman Route Planner with a Blackjack Mini-game

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THE PIZZO

In Italian mafia culture, pizzo means protection money

Toni must visit every business in the city

Streets are dangerous, fuel is expensive

Every unnecessary step increases risk...

Toni MUST choose the smartest possible route

NEIGHBORHOOD

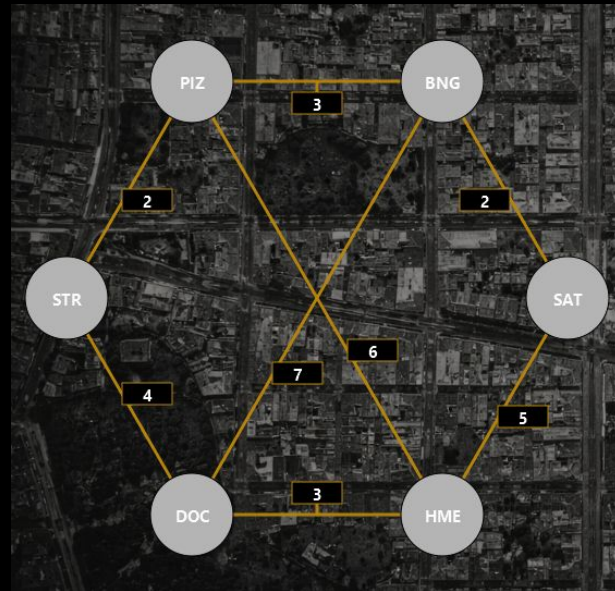
Intersections → vertices

Streets → weighted edges

Cost = distance and risk

Goal:

- **Traverse every street at least once**
- **Return to the starting point**
- **Minimize total cost**



Every Move Has Consequences

First time on a street → earn money

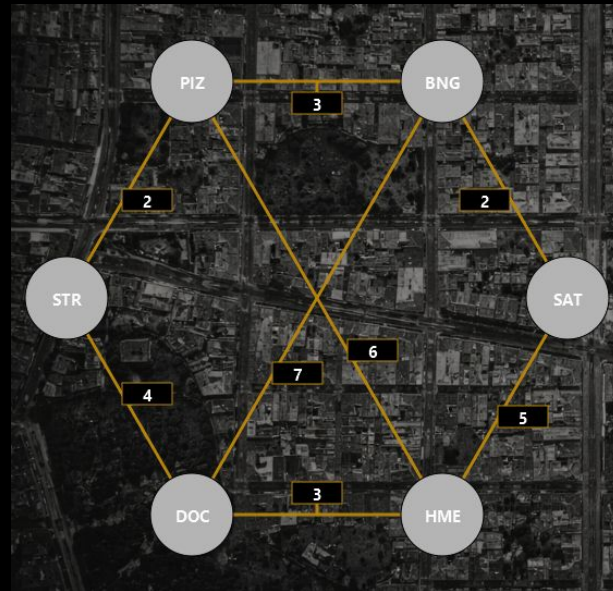
Every traversal → pay a cost

Repeated streets = penalties

Result:

● Smart routes = profit

● Bad routes = losses



The Brain Behind the Wheel

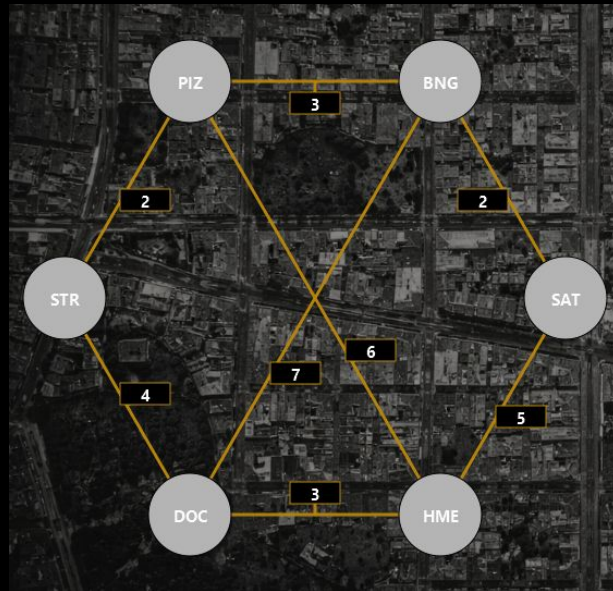
This is the classical Chinese Postman Problem

It solves:

- Which streets must be repeated
- Which must be visited only once
- How to build the shortest possible full tour

High-level steps:

1. Find odd-degree intersections
2. Compute shortest paths between them
3. Pair them intelligently
4. Create a route that becomes perfectly traversable



After Work: Blackjack

Standard 21 rules

Hit or Stand

Dealer follows fixed strategy



The Consigliere

An intelligent advisor calculates:

- Expected value of Hit
- Expected value of Stand

Using:

- Probability model
- Future outcomes
- Optimal decision logic (expectimax)



More than a Game

- The Chinese Postman Problem ensures every edge is covered with minimum cost.
- Odd vertices determine the need for augmentation.
- Perfect matching minimizes duplication cost.
- Hierholzer guarantees a valid Euler circuit.
- Expected Value reasoning guides optimal Blackjack play.



**THANK
YOU!**